
OPERATION, TROUBLESHOOTING AND MAINTENANCE MANUAL

OF MICROPROCESSOR BASED CONTROL AND FAULT DIAGNOSTIC SYSTEM TYPE MCS657

For Electric Locomotives type WAG7, WAG5, WAM4, and WAP4



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1.0 Introduction

Medha's first model of Microprocessor based Control and Fault Diagnostic System Type MCS654 was designed to work with WAG7 and WAP4 class Locomotives in accordance with CLW Specification No. CLW/ES/M-5 in harsh environment of Locomotive on Indian Railways. The equipment has been proven in the field for more than 5 years on Locomotives of Indian Railways, one of the largest Railway network in the world. At present 10 systems of type MCS654 are in service on Locomotives of Indian Railways. This system was designed to perform the logical control of the Locomotive by continuously monitoring various digital and analog inputs e.g. Switches, Feedback contacts, etc and drive the EM Contactors, Valves, etc of the Locomotive. The system was continuously checking for abnormality in the operation. If found, announce the fault condition on the Display Unit mounted in each CAB. This system was equipped with RTC and a Non-volatile memory to record the faults in real time with time and fault type. The Fault Data was down loadable to a PC / Laptop from the system with data extracting software running on PC / Laptop.

With this background, Medha has developed a new Microprocessor based Control and Fault Diagnostic System Type MCS657 to work with WAG7, WAG5, WAM4 and WAP4 class Locomotives fitted with ARNO / Static Inverter and also vertical or roof mounted DBR in accordance with RDSO Specification No. ELRS/SPEC/MPC-FDS/0001 (Rev.-2) Aug 2005. As per this specification the Locomotive type information is read as digital inputs and compared with configured data available in internal memory of unit at power ON, if both are matched then only loco can be charged or it gives warning / error message on display unit. Here also the System continuously monitors various digital and analog inputs e.g. Switches, Feedback contacts, TM Currents, Analog Voltages etc to drive the EM Contactors, Valves, etc of the Locomotive as per the predefined control logic programmed in the Microprocessor of the system. The current status of working locomotive and fault conditions occurred while locomotive working are displayed on the Display Unit mounted in each CAB. The system is equipped with real time clock (RTC) and flash memory to record the faults occurred in real time with date, time and fault type along with background data of all 128 inputs, 80 outputs, Output feed backs, Eliminated Relay status, 6 TM currents and analog inputs, 5 seconds prior to fault and 3 seconds after the fault other than the occurrence of fault (total 9 seconds of data). The Fault Data can be downloaded to a PC / Laptop / Pen drive (Data extraction dongle) through a USB connector provided on fascia of IF card, in control unit of the system. Using data extracting software supplied along with the system running on PC / Laptop, this data can be analyzed by the maintenance staff to troubleshoot / rectify the problem. The Display Unit of the system is provided with a Keyboard and a Menu driven software through which the user can view 'ONLINE" status of various Digital Inputs, outputs, Analog Signals, Eliminated Relay Status, and Diagnostic information, logged faults, Test mode, etc at BLDJ OFF.

The system level block diagram of MCS657 is shown in figure 1.

2.0 Block Diagram of MCS657

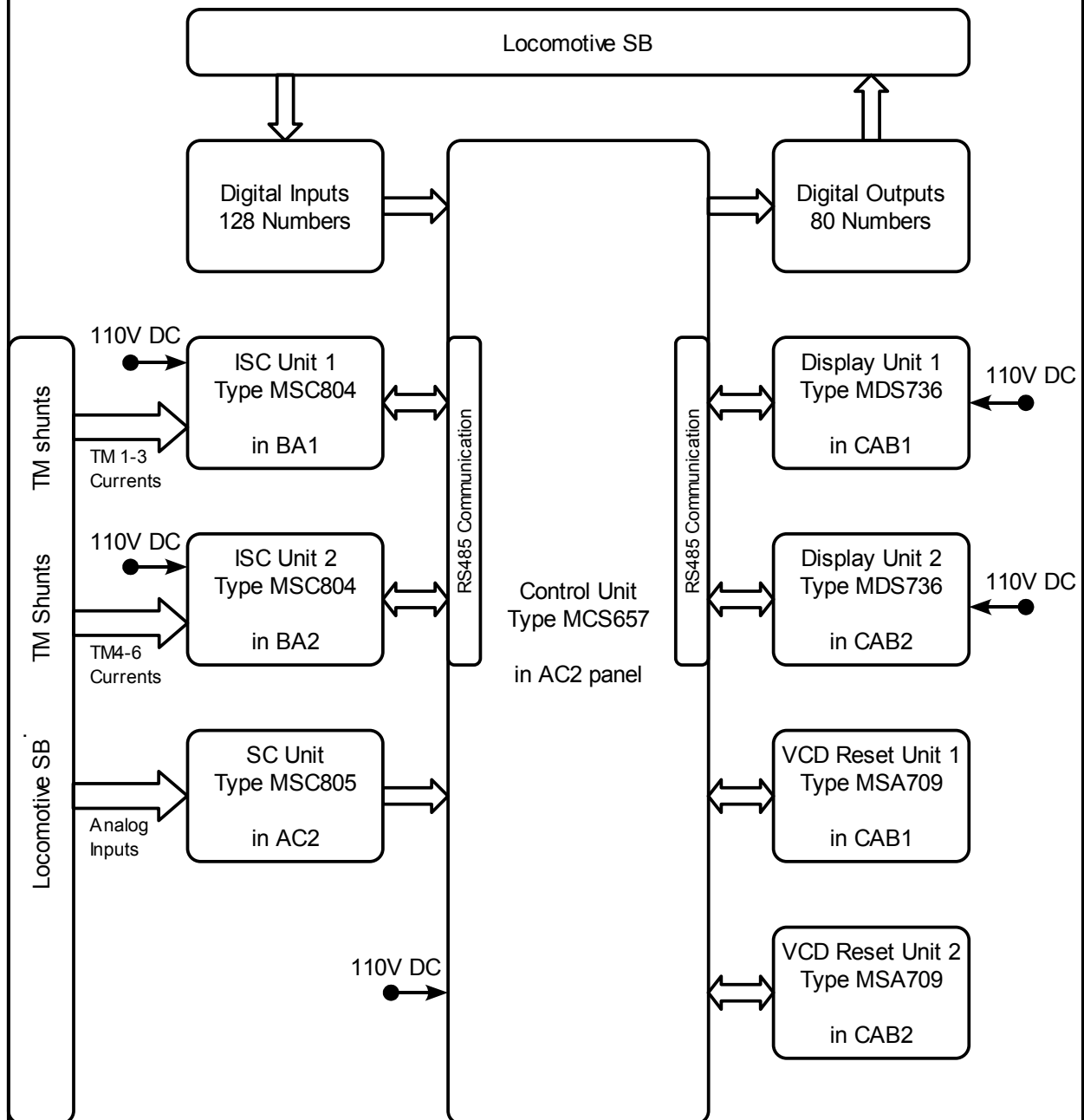


Figure 1

Note: VCD reset unit is optional accessory.

3.0 System Sub Assemblies

The block diagram shown in figure 1 represents the complete system of MCS657. It consists of following sub assemblies.

- | | |
|--|---------|
| 1. Control Unit. | 01 no. |
| 2. Signal Conditioning Unit | 01 no. |
| 3. Intelligent Signal Conditioning Unit. | 02 nos. |
| 4. Display Unit. | 02 nos. |
| 5. VCD Reset unit (Optional) | 02 nos. |

3.1 Control Unit:

Annexure I shows the overall dimensions of the unit. This unit is enclosed in a powder coated MS enclosure and consists of 19 cards in total and necessary Bayonet connectors for all I/Os, communication and power connections. The details of 19 cards are

- | | | |
|-------------------------|---|-------|
| 1. Digital Input Cards | - | 8 nos |
| 2. Digital Output Cards | - | 5 nos |
| 3. Analog Input Card | - | 1 no |
| 4. Interface Card | - | 1 no |
| 5. CPU Cards | - | 2 nos |
| 6. Power Supply Cards | - | 2 nos |

3.1.1 Digital Input Card (MIDIP)

This card consists of the hardware required for converting the high voltage signals of nominal 110V DC from the locomotive circuits to isolated low voltage signals of 5V DC required for the micro processor system. This Card provides 1.5 KV electrical isolation, between locomotive circuits and micro processor circuits. All the inputs are reverse polarity protected, and surge protected. Each card can process 16 input signals. The Green colour LEDs provided on this card fascia indicates presence of 110 V DC voltage on that particular channel. The Yellow LEDs indicate the status of the signal read by Micro processor after isolation and signal conditioning. Thus at any stage, both the LEDs of a particular channel should be either OFF or ON to indicate the correct functioning of card.

8 Digital input cards include one Redundant and one Spare. These 8 cards are interchangeable amongst them.

3.1.2 Digital Output Card(MIHSD)

This card converts 5V digital signals generated by the micro processor system to 110V DC power signals required for driving relays, contactors, indication lamps etc. in the locomotive circuit. This card provides 1.5 KV electrical isolation, between locomotive circuits and micro processor system circuits. Each card can process 16 output signals. This card drives the load connected to it with battery Positive when switched ON. Each output is reverse polarity protected, surge, over load and short circuit protected. The Yellow LEDs indicate the status of the signal given by Micro processor. The Green colour LEDs provided on this card fascia indicates presence of 110 V DC voltage on that

particular channel. Thus at any stage, both the LEDs of a particular channel should be either OFF or ON to indicate the correct functioning of card.

5 Digital output cards include one Redundant and one Spare. These 5 cards are interchangeable amongst them.

3.1.3 Analog Input Card (MCSAN)

This card accepts low voltage Analog signals given by the Signal Conditioning Unit. These signals are isolated and then taken to the CPU card for processing. This card has provision to process up to 8 channels, now 5 channels are in use. Each channel is provided with an LED on fascia for indication. The rate of blinking of the LED indicates the level of the input voltage given to SCU.

3.1.4 IF Card (Interface Card) (MCSIF)

This card communicates with both Display Units. The communication signals are also optically isolated from the rest of the circuit. The communication signal lines are protected by Surge Absorbers. The card communicates with both Display units on RS485 communication standard. This card is also provided with Non Volatile Memory and RTC. The Fault Data is recorded in the Non Volatile read write memory with date, time and type of fault. It has a capacity to store upto 15000 faults and 3000 events data.

This Card is provided with type A USB connector for down loading Fault data to the Pen Drive and type B miniature connector for configuration of various parameters as well as down loading fault data directly on to PC / Laptop. An USB LED is provided on the fascia of this card to know the status of working with Pen drive. Another CS (Card status) LED is provided and it's blinking indicates healthiness of the card. A WD (watch dog) LED on fascia glowing indicates failure abnormality of CPU card.

3.1.5 CPU Card (MCSCC)

The CPU card is designed with 16 bit microcontroller. There are 2 identical CPU Cards. At any time only one CPU Card is active and other remains in stand by mode. This card is also provided with **Watch Dog Timer Circuit** to detect failure of microcontroller or crashing of software. When such a condition occurs the second micro controller takes over automatically and the system can be still run. The LED provided on CPU card blinks whenever it is active. The LED blinking card is in working condition and another is at stand by. This card also communicates with both Intelligent Signal Conditioning Units. The communication signals are again optically isolated from the rest of the circuit. The communication signal lines are protected by Surge Absorbers. The communication with Intelligent Signal Conditioning unit is done on RS485 communication standard. The CPU card reads analog voltages (in the form of frequency) from Signal Conditioning Unit. The CPU card reads the Configuration information from the IF card which can be set by the Laptop. It also sends the status of digital inputs, digital outputs, Output feedback, Eliminated relay status, Analog voltages, Traction motor currents, diagnosis information, notch position and Fault information to the IF card for event logging purpose in case of any fault generated.

3.1.6 Power Supply Card (MCSPS)

This module accepts Locomotive battery supply voltage from 60 to 140 V DC and supplies regulated low voltage DC for the various cards. The power supply is protected from reverse polarity, surges, RFI and EMI through suitable devices. Another card of same type is there in system to take over if the one power supply fails. The following voltages are generated in power supply card and each output is having LED indication.

+9V _I	Isolated DC Supply for the control circuit
+12V _E	Isolated DC Supply for Digital Output Card
+18V _E	Isolated DC Supply for Signal Conditioning Unit
-18V _E	Isolated DC Supply for Signal Conditioning Unit
+9V _E	Isolated DC Supply Communication Circuit

3.2 Signal Conditioning Unit (MSC805)

This unit consists of 5 cards and necessary Bayonet connectors for sending of low voltage signals to control unit. The details of 5 cards are

1. AC Voltage processing cards - 3 nos
2. DC Voltage processing cards - 2 nos

all the above cards with in a group are interchangeable for easy of maintenance.

Annexure II shows the overall Dimensions of the Unit. This unit is enclosed in a Powder Coated MS enclosure. It accepts Locomotive HV Inputs of Auxiliary Supply, ARNO / SI output, Battery Voltage and Traction Motor Voltage (of only one bogie) through a Terminal Block. It outputs isolated Low Voltage signals for further processing by Control Unit through a 10 pin Bayonet Connector. This unit includes one redundant channel for Auxiliary Supply.

3.3 Intelligent Signal Conditioning Unit (MSC804)

This unit consists of two cards and necessary Bayonet connectors for communication and power connections. The details of two cards are

One is Power supply card for converting locomotive's 110 V DC to low level working voltage for the other cards

Second is Control card with it's three child cards for processing of individual TM currents per ISCU. Two ISCU's are required for one system / locomotive to process 6 TM currents.

Annexure III shows the overall Dimensions of the Unit. This unit is enclosed in a Powder Coated MS enclosure. One each unit is mounted in BA1 and BA2 panels. TM Shunt voltages are inputs to this unit. Each unit receives 3 TM Shunt voltages and processed output is sent to controller unit via RS485 communication link. The pick up and drop out currents for auto-regression due to wheel slip can be set through configuration settings in the control unit. This unit also needs 110V DC Locomotive Battery Power Supply. The power supply input is protected from reverse polarity, surges, RFI and EMI through suitable devices.

3.4 Display Unit (MDS736)

The Annexure IV shows the overall dimensions of the unit. This unit is enclosed in a powder coated MS enclosure. The Unit is mounted in the each CAB of the Locomotive. The unit has a 40 character by 4 line back lit type LCD, a 2 digit 7 segment LED display of 1" height and a 8 keys keyboard with hooter. This unit receives data on the Serial RS485 standard communication from Control Unit. The default LCD screen shows date, time and loco configuration settings. The 7-segment display shows current notch number. Any Fault is announced with a Fault Message on LCD with hooter (If configured to announce). The LCD displays the Current Status, Earlier Logged Faults, Isolation conditions, Digital and Analog Inputs, Digital Outputs, etc depending on the option selected through the keyboard. This unit takes 110 V DC Locomotive Battery Power Supply for providing various regulated DC low voltages for functioning of Display Unit. The power supply is protected for reverse polarity, surges, RFI / EMI, under and over voltage of supply through suitable devices.

4.0 Principle of Operation

Control Unit:

After the power is switched ON to the Control Unit, Display Unit and ISCU, one of the CPU cards becomes active, it checks for healthiness of various sub systems. Any abnormality found is sent to both Display Units for announcement. The Control Unit reads the Digital and Analog input signals and debounces them through software. Depending on the state of various inputs logical condition of Digital Outputs is worked out based on the predefined logic. Digital outputs are set to high (+ve of Battery supply voltage) or Low (Off) state through the Digital output cards. Also the initial status of certain inputs e.g Reverser is checked at BL key switch ON. Any abnormality is immediately announced on display Units. The monitoring of inputs and calculation of outputs is done at a very high rate to ensure a prompt action. If any feedback input does not match with the output driving status during running of locomotive, then also an appropriate message is sent to Display Units. If the set limit of TM voltage, OHE voltage, ARNO voltage etc.. is crossed, then either Auto regression or DJ tripping is done or any other appropriate action is taken as per the defined logic. Faults are logged in a Non Volatile Memory with date, time and type of fault and background data. When the fault data is to be downloaded, the USB port of PC / Laptop / Pen drive is connected to the USB connector of Interface card. The downloaded data can be analyzed using the data extraction and analysis software on the PC / Laptop. For every normal DJ opening with BLDJ, the standby CPU card becomes active and active CPU card becomes standby. If at any stage active CPU card fails, the other healthy CPU card takes over with a message on Display Units and logging of fault. Once a CPU cards is found faulty the role change over does not take place. The system works only with the healthy CPU. If both the CPU cards fails, then all the outputs will go into safe mode i.e. de energised condition (0V).

Display Unit:

On powering up, each Display Unit checks for communication from Control Unit. If no communication is received within 10 seconds the respective Display announces a communication fail message. The system can continue to work even if one or both Display Units fail. The default screen

on the LCD shows Date, Time and loco configuration which was set. The 2 digit 7 segment LED indicator shows the current notch position. When Control Unit is unable to find out the current notch position e.g. after power up of any the control Unit the Tap Changer is not at '0', Manual Operation of GR, etc, the display shows "Er". Any fault message received from Control Unit is announced by Displaying of Message on LCD and Sounding of Hooter (If configured). For messages which need compulsory acknowledgment, the hooter continues to sound till acknowledge key is pressed in the CAB where BL key is made 'ON'. For other messages and Display in the Other CAB, the Hooter stops sounding after 10 seconds irrespective of the type of message.

The Keyboard provided on the Display Unit can be used to get the information and current working status of the system. There are 8 Membrane keys. These are "MENU", "▲", "▼", "ENTER", "ACK", Back light ON / OFF, Contrast increase and Contrast decrease.

Pressing the "MENU" Key shows the following Screen

Date: 04-10-06	/ 1. Vehicle Diagnostic	Time 11:12:13
	2. Process Information	
	3. Exit	

The "/" shows the current cursor position. To select an option, move cursor up or down by using "▲", "▼" keys. To select the option, press "ENTER" key. The next screen for "Vehicle Diagnostic" menu is as follows

Date: 04-10-06	/ 1. Status Information	Time 11:12:13
	2. Fault Information	

On the above screen, if we press "ENTER" key again, the next screen of status information is displayed. In this screen we have isolation Information about HVMT1, HVMT2 and HVRH. Or if we press "▼" key and "ENTER" then the display enters into Fault information screen as follows

Date: 04-10-06	/ 1. Active Faults	Time 11:12:13
	2. Logged Faults	
	3. Erase Active Faults	

In the above screen Active Faults are the faults which are active and are to be attended. After rectifying the corresponding faults, we can erase them using above Erase Active Faults option. Here the fault message only displayed like

Active Faults

DJ tripped Via QOP-1
Press '↓' or '↑' Key to See Other Faults

And Logged Faults are those stored in Non volatile memory. With the fault, date and time of the fault occurrence will be displayed. We can go back and forth to see other faults.

In Process Information we have options like

- / 1. Input / Output Display
- 2. Parameter Setting
- 3. Test Mode

The following Screen is shown on selection of "Input/Output Display"

- / 1. Display Digital Input
- 2. Display Digital Output
- 3. Eliminated Relay Status
- 4. Display Analog Parameters

The following Information is shown on selection of Display Digital Input option

Input Card 1		I0 to I7
BLDJ = 1	BLRDJ = 0	QVMT1 = 1
QVMT2 = 1	QVRH = 1	ZPT 1 = 1
ZPT 2 = 0	HVMT11 = 1	↓

The first input starts with I0 i.e BLDJ. In one screen 8 digital inputs status can be seen. The status of input "0" shows that it is in Low state (OFF, "0" volts), whereas, the status "1" shows that it is in High state (on, 110 VDC). In this way all the 128 Digital Inputs are shown in sequence. We can go back and forth for the other screens. Like this for all Digital outputs also there are separate screens with "0" means output is in Low state (OFF, "0" volts), whereas a "1" indicates that it is in High state (ON, "110" volts DC). In this way all the 80 Digital Outputs are shown in sequence.

On selection of Eliminated Relay status, a "0" means that the calculated status as per conventional loco logic based on inputs is "OFF" state. "1" means that the calculated status as per conventional loco logic based on inputs is "ON"

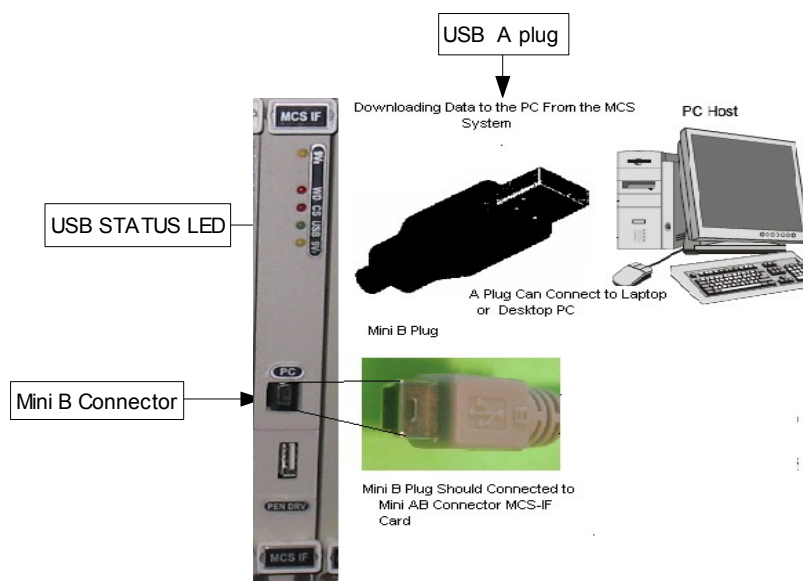
5.0 USB Communication

Two ways of Downloading Data from the System is possible.

1. MCS657 System to the Laptop / Desktop
2. MCS657 System to the USB Memory Stick (PEN DRIVE)

Downloading Data to the Laptop / Desktop

1. The Operating system should be windows 2000 or above.
2. User need to use USB A-to mini B cable for Downloading data to Desktop / Laptop. Mini-B Plug should be connected to MCS657 System and USB-A plug should be connected to Laptop or Desktop.
3. User need to ensure that Loco is in denergised Condition by BLDJ OPEN and GR at 0 before going for downloading operation of Pen drive or laptop. MCS657 System allows user for USB communication only at Loco denergised condition. System will announce a message **Ensure Loco Standstill condition** on both display units when Laptop / Desktop PC is connected at Loco energised condition.
4. User can download data from the MCS657 system and analyse data by using Data Analysis Software.
5. USB Status LED (on IF card) toggling indicates the communication between PC and MCS657 System is in progress.
6. User can also configure MCS657 System parameters by using Data analysis Software.
7. After downloading or configuring operation user should disconnect USB A- to Mini B cable from MCS657 System for further system operation



Downloading Data to the PENDRIVE (USB MEMORY STICK)

1. User need to Connect USB Memory Stick (PEN DRIVE) to USB-A Connector of MCS657 System.
2. User need to ensure that Loco is in denergised Condition by BLDJ OPEN and GR at 0 before going for Downloading Operation of Pen drive. MCS657 System will allow user for USB communication only at Loco denergised condition. System will announce **Ensure Loco Stand still condition** in Display when Pen drive is connected at Loco Running Condition.
3. User should Ensures that Pen drive is not write protected.
4. MCS657 System will automatically download the data to the pen drive when it is connected and Loco is in Standstill condition.
5. USB Status LED toggling indicates the Communication between MCS657 System and Pen drive. USB Status LED will stop it's blinking when it completes USB related operation. Do not remove Pen Drive until the LED Stops blinking. If removed while communication is in progress may corrupt the Pen Drive.
6. After downloading all the data display will give message that Files downloaded Successfully.

6.0 Troubleshooting Guide Lines

The System is built with no moving parts or contacts, hence there is no periodic maintenance required. How ever there can problems of following nature

- a). A Digital Input or Digital Output is not at the desired level due to faults in the Digital Input / Digital Output card.
- b). Faults due to faulty cable / interconnections between various units of the system
- c). Power Supply card failure resulting in "DEAD" the system

We recommend only card level maintenance by Shed staff for any problem with the cards. After opening the front door of Control Unit by Unlocking the Key, Glowing of the LEDs provided on different cards can be used to check for working of the cards as explained below

Nature of Problem	Check for	Action
Digital Input not at Desired Level	1.Voltage level on terminal SB 2.MG Connector's proper connection. 3.Corresponding Input and Output LEDs of the Digital Input Card 4.Digital input status on LCD of Display Unit for the corresponding Input	Change the corresponding input card when the Voltage is available on SB and not coming on LCD. or If LEDs "Yellow and Green" on fascia do not match.
Digital Output not at Desired	1.Digital output status on LCD	If Terminal SB Voltage and

Nature of Problem	Check for	Action
Level	of Display Unit for the corresponding output 2.Voltage level on Terminal SB 3.Load short circuited with Locomotive body / B - .	displayed information on LCD does not match Change corresponding Digital Output Card
Traction motor voltage or battery voltage not coming	Confirm the Voltage up to SCU input terminals. Interconnections between Signal Conditioning Unit and Control Unit Display on LCD for Analog voltages Display	Rectify cable if found faulty If voltage coming upto SCU Terminal SB and not coming on display, then either Signal Conditioning Unit or Analog Input may be faulty. Identify the problem area and change SC Unit or Analog Input Card
Auxiliary Voltage is not coming or ICDJ through OHE low message on display while closing the DJ	Confirm the Voltage up to SCU input terminals. Display on LCD for Analog voltages Display	If voltage at SCU terminal block and not displaying on display unit when closing the DJ, then replace the corresponding MCSCAC card in the SCU or MCSAN
Control Unit Dead	Check the fuse on control unit Check PS cable Check LED on PS Card	IF cable and 110 Vdc Supply Voltage is present. PS Card may be faulty if all LEDS do not glow. Change PS card
Display Unit Dead	Check presence of 110 Vdc PS No display on the Display Unit	If PS is present Display Unit may be faulty. Change Unit
Both or One Display Communication Fail	Check Cable between Display and Control Unit	If Cable is OK. Either Display Unit or Interface Card May be Faulty. Change Card.

7.0 Maintenance Schedule

There is no need of periodical maintenance for microprocessor based control and fault diagnostic system, because there are no moving parts in the system and hence there are no periodical replaceable components.

As a safety and preventive actions the following checks can be carried out when the loco comes for schedules.

1. Check the fault log through Display unit / laptop and do the necessary action according to the fault log.
2. Check display units in both cabs and its operation, through menu driven keys.
3. Check the proper tightening of Bayonet connectors at Control unit, Signal conditioning unit,

Intelligent Signal Conditioning Unit, Display units and control cable termination on loco terminal blocks.

4. Perform locomotive functional testing as per schedule.

8.0 Specifications

Control Unit

Supply Voltage	60V to 140V DC
Power consumption	25W approx
Suitable for locomotives	WAG5, WAG7, WAP4 and WAM4
I/O couplers	Reverse Bayonet connectors
Number of Digital Inputs	128 Nos. with LED indication on fascia
Number of Digital Outputs	80 Nos. with LED indication on fascia
Number of Analog Inputs	5 Nos. provision for 8 nos with LED indication on fascia
Number of TM Current Inputs	6 Nos.
Dimensions	As per Annexure I
Weight	55 Kgs approximately

Display Unit

Supply Voltage	60V to 140V DC
Power consumption	10W approximately
I/O couplers	Reverse Bayonet connectors
LCD	40 X 4 line alphanumeric with back lit provision
Notch Indicator	Two digit 7 segment indication
Back lit	ON / OFF control
Contrast	Adjustable through Keyboard
Key board	8 keys of sealed membrane key board
Dimensions	As per Annexure IV
Weight	2 Kgs approximately

Intelligent Signal Conditioning Unit

Supply Voltage	60V to 140V DC
Power consumption	10W approx
I/O couplers	Reverse Bayonet connectors
Number of TM Current Inputs	3 Nos.
Dimensions	As per Annexure III
Weight	3 Kgs. Approximately

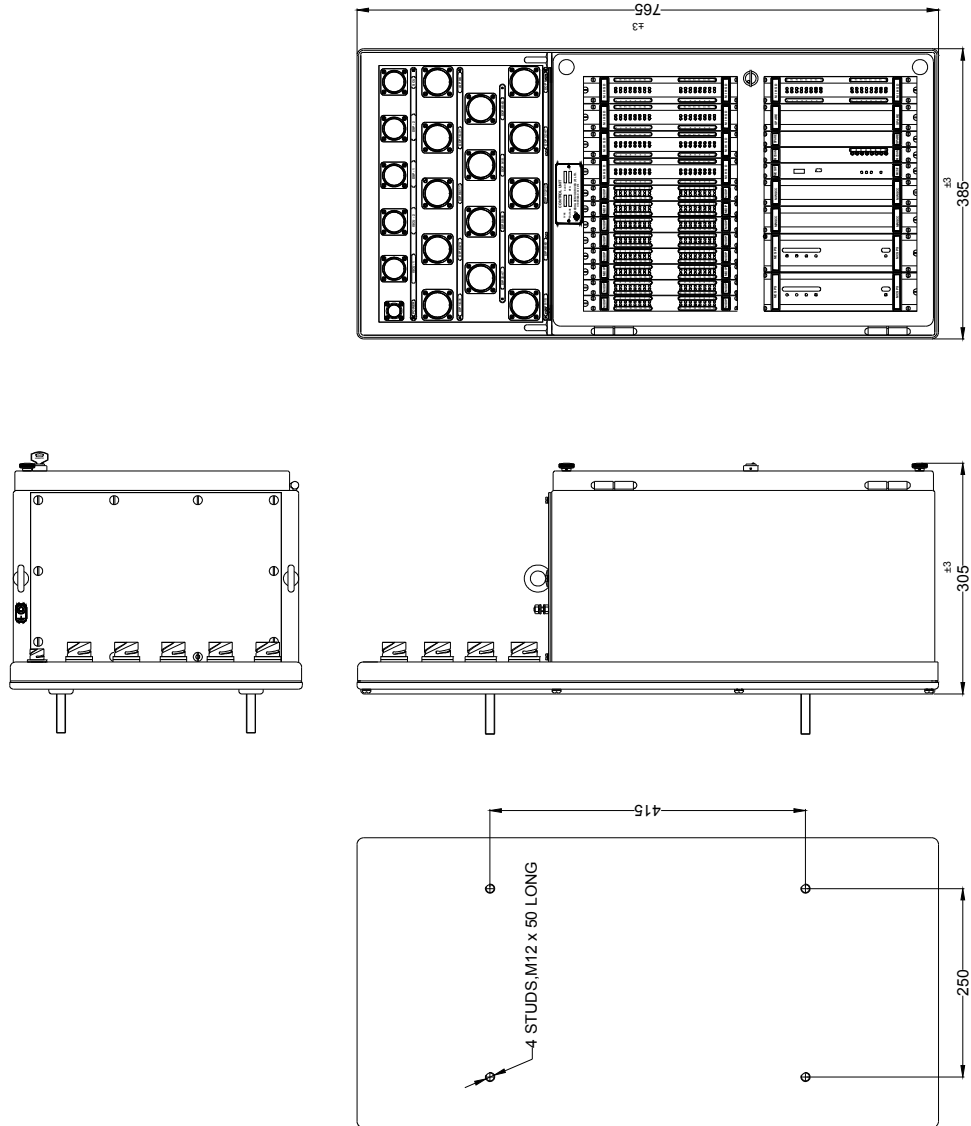
Signal Conditioning Unit

Supply Voltage	18V DC from Control Unit
Power consumption	10W approx
I/O coupler	Reverse Bayonet connector
Number of Analog voltage channels	5 Nos.(3 for AC voltage and 2 for DC voltage)
Dimensions	As per Annexure II
Weight	3 Kgs. approximately

Abbreviations used :

CU	Control Unit
SCU	Signal Conditioning Unit
ISCU	Intelligent Signal conditioning Unit
CPU	Central Processing Unit
LED	Light Emitting Diode
LCD	Liquid Crystal Display
USB	Universal serial Bus
IF	Interface
ARNO	Single phase to 3 phase AC converter
SI	Static Inverter
TM	Traction Motor
RTC	Real time Clock
PC	Personal Computer

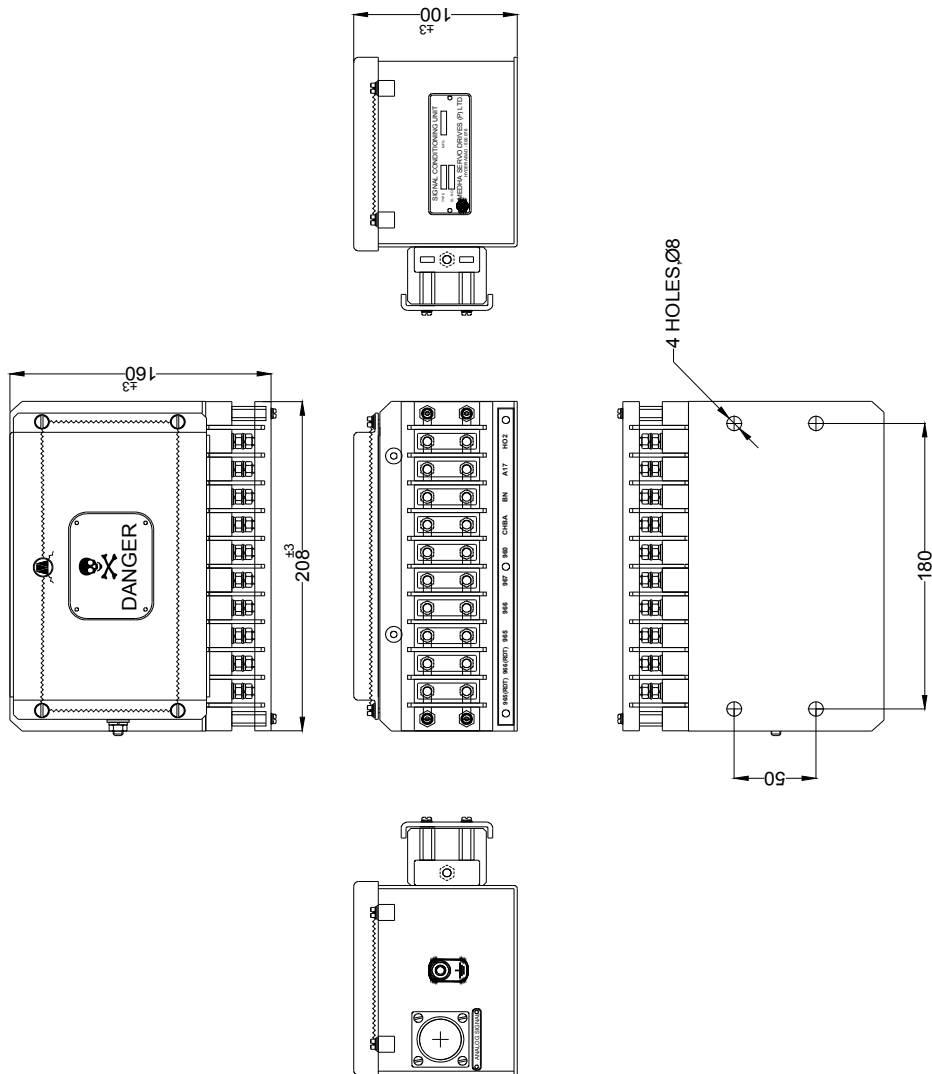
Annexure I



CONTROL UNIT ASSY TYPE MCS 657 TO
DRG. NO. B - C657 - 6419, REV. NO. : 00

ALL DIMENSIONS ARE IN MM

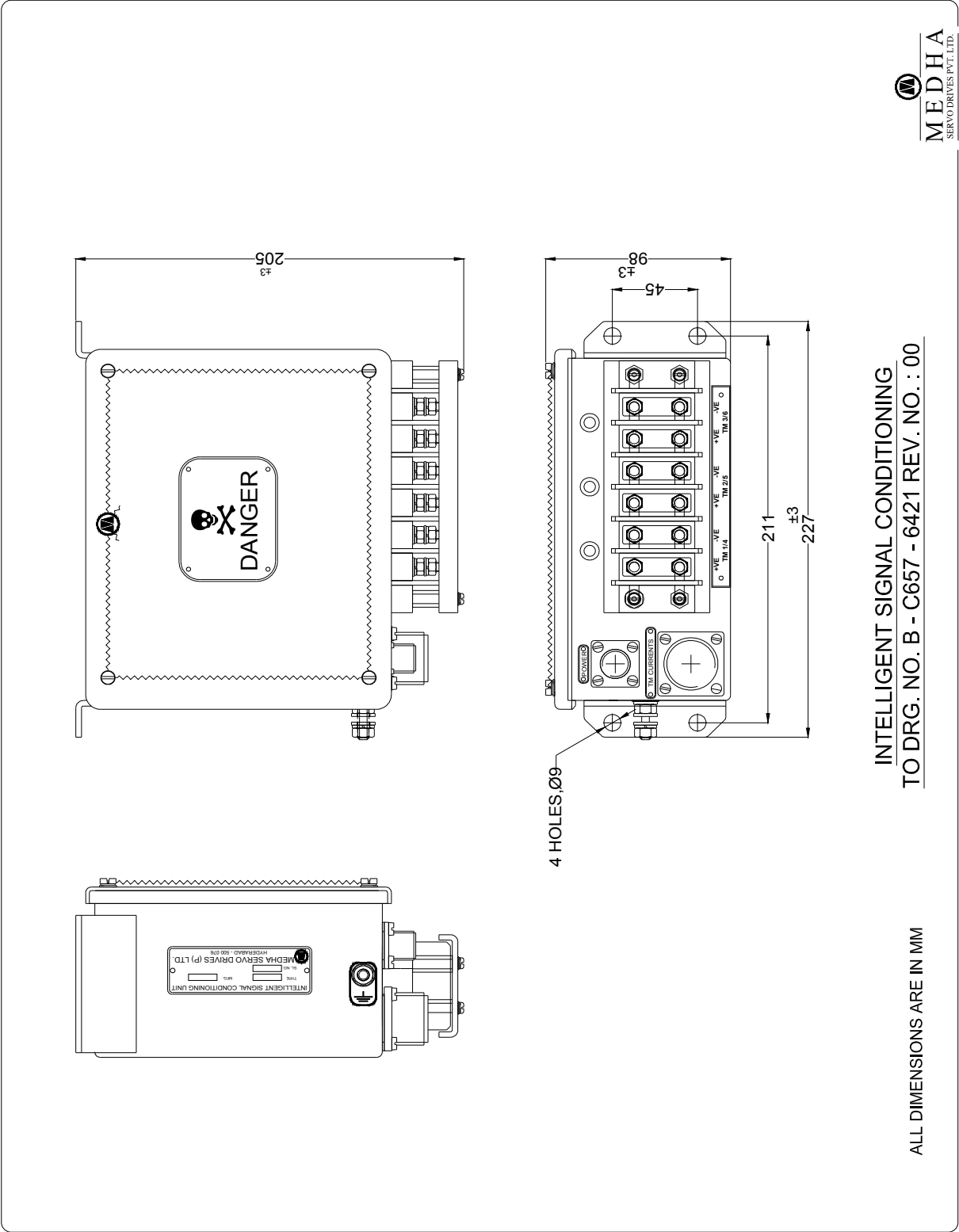
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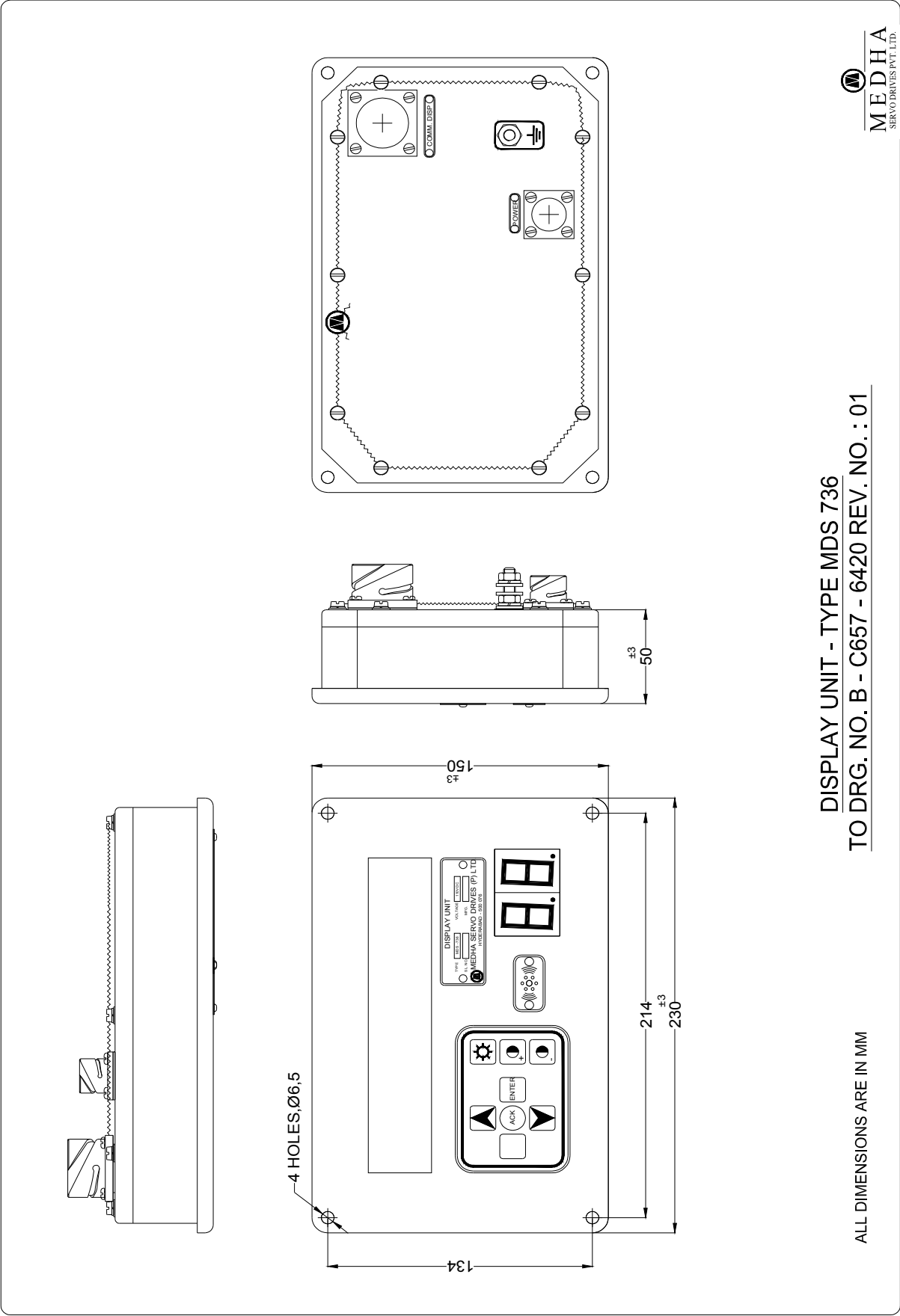
SIGNAL CONDITIONING UNIT
TO DRG. NO. B - C657 - 6422 REV. NO. : 00

ALL DIMENSIONS ARE IN MM

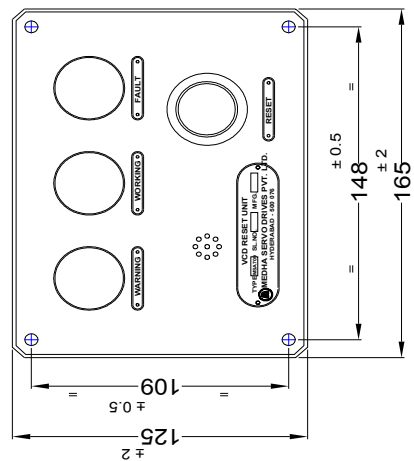
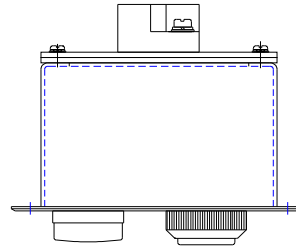
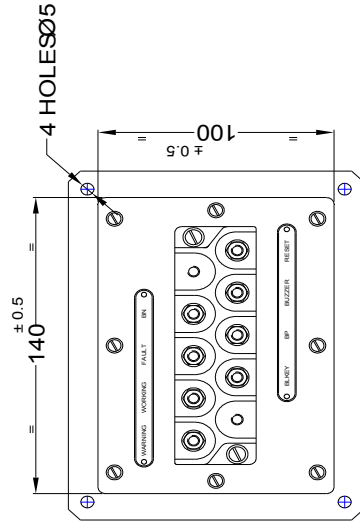
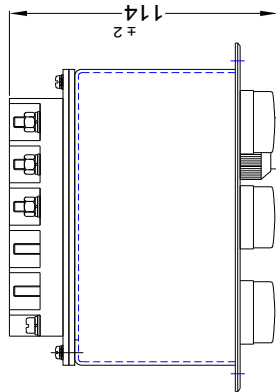
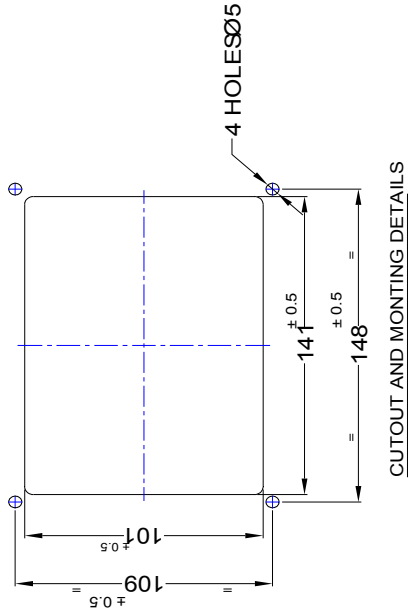
Annexure III



Annexure IV



Annexure V



NOTE : ALL DIMENSIONS ARE IN "mm"

Annexure VI
Digital Input Details

Card No	Bayonet Connector		Signal		Ferrule on Core
	Type & Ferrule	Pin No	Legend	Name	
1	MG06F 22-14S I0 - I15	A	I - 0	BP1DJ / BLDJ	021
		B	I - 1	BP2DJ / BLRDJ	024
		C	I - 2	QVTM1	025
		D	I - 3	QVTM2	026
		E	I - 4	QVRH	027
		F	I - 5	ZPT1_2	030
		G	I - 6	ZPT1_1	029
		H	I - 7	HVMT1_1	036
		J	I - 8	HVMT1_2	037
		K	I - 9	HVMT2_1	038
		L	I - 10	HVMT2_2	039
		M	I - 11	HVRH_1	040
		N	I - 12	HVRH_2	041
		P	I - 13	QVSI1 / HVSI1	042
		R	I - 14	QVSI2 / HVSI2	043
		S	I - 15	BLVMT	070
		T		NC	NC
		U		NC	NC
		V		BN	BN

Card No	Bayonet Connector		Signal		Ferrule on Core
	Type & Ferrule	Pin No	Legend	Name	
2	MG06F 22-14SW I16 - I31	A	I - 16	ASMGR (BN)	072
		B	I - 17	BLCP / BLCPD	074
		C	I - 18	C101_3FB	075
		D	I - 19	GR - 0	076
		E	I - 20	GR - 0_31	077
		F	I - 21	QPH / HPH	078
		G	I - 22	QVSL1 / HVSL1	079
		H	I - 23	C105_FB	061
		J	I - 24	ASMGR (ON)	082
		K	I - 25	QVSL2 / HVSL2	080
		L	I - 26	MP+ (R,B)	093
		M	I - 27	MPJ (FOR)	091
		N	I - 28	J1,J2 (FOR)	095
		P	I - 29	MP- (R,B)	096
		R	I - 30	MP (+,N,-) R	097
		S	I - 31	CTF (RUN)	100
		T		NC	NC
		U		NC	NC
		V		BN	BN

Card No	Bayonet Connector		Signal		Ferrule on Core
	Type & Ferrule	Pin No	Legend	Name	
3	MG06F 22-14SX I32 - I47	A	I - 32	MPJ (REV)	092
		B	I - 33	CTF (BRK)	212
		C	I - 34	MP (+,N,-) B	213
		D	I - 35	DJ_FB	105
		E	I - 36	J1,J2 (REV)	107
		F	I - 37	ZQWC	121
		G	I - 38	MPS (1-4)	123
		H	I - 39	MPS (2-4)	124
		J	I - 40	MPS (3-4)	125
		K	I - 41	MPS 4	126
		L	I - 42	PVEF	150
		M	I - 43	PSA	151
		N	I - 44	BPQD	230
		P	I - 45	RGEB	155
		R	I - 46	SWC	156
		S	I - 47	QF & QE	162
		T		NC	NC
		U		NC	NC
		V		BN	BN

Card No	Bayonet Connector		Signal		Ferrule on Core
	Type & Ferrule	Pin No	Legend	Name	
4	MG06F 22-14SY I48 - I63	A	I - 48	BL	142
		B	I - 49	QOA / QSIT	045
		C	I - 50	QOP1	046
		D	I - 51	QOP2	047
		E	I - 52	QRSI1	048
		F	I - 53	QRSI2	049
		G	I - 54	QLM	050
		H	I - 55	BV	122
		J	I - 56	C106_FB	062
		K	I - 57	ZSMGR	128
		L	I - 58	ZSMS	120
		M	I - 59	RSI	170
		N	I - 60	HMCS	058
		P	I - 61	L1 TO L6_FB	028
		R	I - 62	HMCS & QD	153
		S	I - 63	C118 N/C/QCON	018
		T		NC	NC
		U		NC	NC
		V		BN	BN

Card No	Bayonet Connector		Signal		Ferrule on Core
	Type & Ferrule	Pin No	Legend	Name	
5	MG06F 22-14SZ I64 - I79	A	I - 64	QLA_FB/SI INT FAULT	019
		B	I - 65	QPDJ_FB	052
		C	I - 66	C107_FB	060
		D	I - 67	SI EXT FAULT	068
		E	I - 68	C145 N/O	059
		F	I - 69		
		G	I - 70	HQ51	200
		H	I - 71	CHBA	973
		J	I - 72	BL1	149
		K	I - 73	SWITI/DBR	157
		L	I - 74		
		M	I - 75	RGAF(ACP)	219
		N	I - 76	P2	216
		P	I - 77	BPT	217
		R	I - 78	RGPA/P1 (ACP)	218
		S	I - 79	BPSW1/2/ACK (ACP)	203
		T		NC	NC
		U		NC	NC
		V		BN	BN

Card No	Bayonet Connector		Signal		Ferrule on Core
	Type & Ferrule	Pin No	Legend	Name	
6	MG06F 20-29S I80 - I95	A	I - 80	LOCO SEL 1	SEL-1
		B	I - 81	LOCO SEL 2	SEL-2
		C	I - 82	LOCO SEL 3	SEL-3
		D	I - 83	LOCO SEL 4	SEL-4
		E	I - 84	MU_FB	
		F	I - 85	ES_DJ	067
		G	I - 86		
		H	I - 87		
		J	I - 88		
		K	I - 89		
		L	I - 90		
		M	I - 91		
		N	I - 92		
		P	I - 93		
		R	I - 94		
		S	I - 95		
		T		BN	BN

Card No	Bayonet Connector		Signal		Ferrule on Core
	Type & Ferrule	Pin No	Legend	Name	
7	MG06F 20-29SW I96 - I111	A	I - 13	QVSI2 / HVSI2	042
		B	I - 21	QPH / HPH	078
		C	I - 22	QVSL1 / HVSL1	079
		D	I - 25	QVSL2 / HVSL2	080
		E	I - 49	QOA / QSIT	045
		F	I - 50	QOP1	046
		G	I - 51	QOP2	047
		H	I - 52	QRSI1	048
		J	I - 53	QRSI2	049
		K	I - 54	QLM	050
		L	I - 64	QLA	051
		M	I - 63	C118 N/C/QCON	059
		N	I - 48	BL	142
		P	I - 15	BLVMT	070
		R	I - 17	BLCPD	074
		S	I - 65		
		T		BN	BN

Card No	Bayonet Connector		Signal		Ferrule on Core
	Type & Ferrule	Pin No	Legend	Name	
8	MG06F 20-29SZ I112 - I127	A	I - 112		
		B	I - 113		
		C	I - 114		
		D	I - 115		
		E	I - 116		
		F	I - 117		
		G	I - 118		
		H	I - 119		
		J	I - 120		
		K	I - 121		
		L	I - 122		
		M	I - 123	HORN	HORN
		N	I - 124	A9	A9
		P	I - 125	SA9	SA9
		R	I - 126	RESET	RESET
		S	I - 127	LOW SPEED	LOW SPEED
		T		BN	BN

Annexure VII
Digital Output Details

Card No	Bayonet Connector		Signal		Ferrule on Core
	Type & Ferrule	Pin No	Legend	Name	
1	MG06F 22-14P O0 – O15	A	O - 0	DJ	044
		B	O - 1	C118 BLSI	035
		C	O - 2	VEPT1	055
		D	O - 3	VEPT2	056
		E	O - 4	DJ	044
		F	O - 5	C107	083
		G	O - 6	C106	084
		H	O - 7	C105	085
		J	O - 8	C101, C103	086
		K	O - 9	VEUL	087
		L	O - 10	J1, J2 (FOR)	108
		M	O - 11	J1, J2 (REV)	109
		N	O - 12	CTF (RUN)	111
		P	O - 13	CTF (BRK)	112
		R	O - 14	VE (UP)	110
		S	O - 15	C145	114
		T		BP	BP
		U		BP	BP
		V		BN	BN

Card No	Bayonet Connector		Signal		Ferrule on Core
	Type & Ferrule	Pin No	Legend	Name	
2	MG06F 22-14PW O16 – O31	A	O - 16	VE (DN)	113
		B	O - 17	EVPHGH	115
		C	O - 18	Sx1	129
		D	O - 19	Sx2	130
		E	O - 20	Sx31	131
		F	O - 21	Sx32	132
		G	O - 22	IP	166
		H	O - 23	VESA2	165
		J	O - 24	VEF	164
		K	O - 25	VESA1	163
		L	O - 26	L1, L2, L3	143
		M	O - 27	L4, L5, L6	133
		N	O - 28	LSDJ (R)	171
		P	O - 29	LSCHBA (G)	172
		R	O - 30	LSGR (G)	173
		S	O - 31	LSB (Y)	174
		T		BP	BP
		U		BP	BP
		V		BN	BN

Card No	Bayonet Connector		Signal		Ferrule on Core
	Type & Ferrule	Pin No	Legend	Name	
3	MG06F 22-14PX O32 – O47	A	O - 32	LSP (R)	175
		B	O - 33	LSRSI (Y)	176
		C	O - 34	SX41	144
		D	O - 35	SX42	145
		E	O - 36	LSGROUP (R)	235
		F	O - 37	LSOL (Y)	210
		G	O - 38	LPAR (R)	179
		H	O - 39	SON (ALARM)	177
		J	O - 40	Q49/MU	
		K	O - 41	B1, B2 (CPA)	231
		L	O - 42	LSDBR (Y)	234
		M	O - 43	LSFL	232
		N	O - 44	QFL	236
		P	O - 45	FL_LP	
		R	O - 46	C102	
		S	O - 47	C108	
		T		BP	BP
		U		BP	BP
		V		BN	BN

Card No	Bayonet Connector		Signal		Ferrule on Core
	Type & Ferrule	Pin No	Legend	Name	
4	MG06F 22-14PY O48 – O63	A	O - 48	DJ	044
		B	O - 49	C118/BLSI	035
		C	O - 50	VEPT1	055
		D	O - 51	VEPT2	056
		E	O - 52	DJ	044
		F	O - 53	C107	083
		G	O - 54	C106	084
		H	O - 55	C105	085
		J	O - 56	C101, C103	086
		K	O - 57	VE (DN)	113
		L	O - 58	J1, J2 (FOR)	108
		M	O - 59	J1, J2 (REV)	109
		N	O - 60	CTF (RUN)	111
		P	O - 61	L1,L2,L3	143
		R	O - 62	VE_UP	110
		S	O - 63	L4,L5,L6	133
		T		BP	BP
		U		BP	BP
		V		BN	BN

Card No	Bayonet Connector		Signal		Ferrule on Core
	Type & Ferrule	Pin No	Legend	Name	
5	MG06F 22-14PZ O64 – O79	A	O - 64		
		B	O - 65		
		C	O - 66		
		D	O - 67		
		E	O - 68		
		F	O - 69		
		G	O - 70		
		H	O - 71		
		J	O - 72		
		K	O - 73		
		L	O - 74		
		M	O - 75	WORKING	WORKING
		N	O - 76	FAULT	FAULT
		P	O - 77	EM VALVE	EM VALVE
		R	O - 78	BUZZER	BUZZER
		S	O - 79	WARNING	WARNING
		T		BP	BP
		U		BP	BP
		V		BN	BN